

Personal repository edition

Updated 20 September 2026: source template, generated target, and companion-repository imports clarified.

Source: https://github.com/alvinea28/own-ml-ops-v2/blob/main/documentation/deployguides/deployguide_ado.md

Original screenshots retain upstream example labels; follow the updated repository names in the text.

Deployment Guide - Azure DevOps

This document will guide you through using the MLOps V2 project generator to deploy a single-environment ("Prod") demo project using only Azure DevOps to host source repositories and pipelines. See notes at the end for guidance on multi-environment MLOps and adapting the pattern to your use case.

Personal repository edition: Use [alvinea28/own-ml-ops-v2](#) on **main**, not the original Azure accelerator. The default classical / AML CLI v2 / Bicep path uses the repaired [taxi project](#). See [the personal setup notes](#) for prerequisites and differences from the upstream examples. Screenshots retain their original example labels; use the repository names and parameters in the updated text below.

What is imported, and what does the initializer create?

The imported taxi repository is a reusable source template, not the new project. The initializer populates a separate target repository with the working code and creates its pipeline definitions. It does not create the Azure Repos repository itself; create that empty target before running the initialization YAML.

Role	Azure Repos repository	How it is created
SOURCE: repaired template	<code>taxi-fare-regression-template</code>	Import alvinea28/taxi-fare-regression from GitHub once. It contains the reusable working fixes.
TARGET: your new project	For example, <code>taxi-fare-regression-demo</code>	Create a new empty Azure Repos repository. The initializer copies the source files into it, commits them, and pushes them. Do not import the taxi template into this target.

Run the initializer with `taxiTemplateRepoName: taxi-fare-regression-template` and `repoName: taxi-fare-regression-demo` (or your own new target name). Never select your existing working taxi project or the source template as the target.

Which repositories are actually in the personal GitHub account? Only [alvinea28/own-ml-ops-v2](#) and [alvinea28/taxi-fare-regression](#) were created for this workflow. The two upstream companion repositories have **not** been copied into that account:

- [Azure/ml-ops-templates](#) remains a **required** shared dependency for the generated training and deployment pipelines.
- [Azure/ml-ops-project-template](#) is **optional for this taxi walkthrough**. It is needed for other scenario selections or the upstream opt-out, not for the repaired classical / AML CLI v2 / Bicep path.

Import sources **from GitHub into Azure Repos**. Even if you later create personal copies of the companions, the current YAML still needs Azure Repos imports because it uses Azure Repos `git://` checkouts, not direct GitHub checkouts. The import table below intentionally keeps the real upstream URLs for dependencies that were not copied.

[PDF version of this guide](#)

Prerequisites:

- One or more Azure subscription(s) based on whether you are deploying Prod only or Prod and Dev environments
 - **Important:** - As mentioned in the **Prerequisites** at the beginning [here](#), if you plan to use either a Free/Trial or similar learning purpose subscriptions, they might pose 'Usage + quotas' limitations in the default Azure region being used for deployment. Please read provided instructions carefully to successfully execute this deployment.
- An Azure DevOps organization
- Ability to create Azure service principals to access / create Azure resources from Azure DevOps
- If using Terraform to create and manage infrastructure from Azure DevOps:
 - Install the [Terraform extension for Azure DevOps](#)
 - **This modernized version requires Terraform v1.10.0 or higher** with azurerm provider ~> 4.52.0
- **Python 3.11 or higher** for local development and testing

Steps to Deploy

1. [Clone and Configure the MLOps V2 Solution Accelerator](#)
Create a copy of the MLOps V2 Solution Accelerator in your organization that can be used to bootstrap new ML projects.
2. [Create and Configure a New ML Project Repo](#)
Use the solution accelerator to create a new ML project according to your scenario and environments and configure it for deployment.
3. [Deploy and Execute Azure Machine Learning Pipelines](#)
Run Azure DevOps pipelines in your new project to deploy Azure Machine Learning infrastructure, deploy and run a training pipeline, and a deployment pipeline.

Clone and Configure the MLOps V2 Solution Accelerator

This section guides you through creating an Azure DevOps project to contain the MLOps repositories and your ML projects, importing the MLOps repositories, and configuring the project with permissions to create new pipelines in the ML projects you generate.

Below are the three repositories to import for the repaired taxi scenario. They serve as the "project factory" for classical / AML CLI v2 / Bicep projects. The upstream project-template repository is optional for other scenarios.

Repository	Azure Repos name	Role
https://github.com/alvinea28/own-ml-ops-v2	<code>mlops-v2</code> in this walkthrough, or <code>own-ml-ops-v2</code> if you keep the GitHub name	Personal accelerator with the repaired initializer on <code>main</code> .
https://github.com/alvinea28/taxi-fare-regression	<code>taxi-fare-regression-template</code>	Repaired standalone taxi project; use a different repository for the generated application.
https://github.com/Azure/ml-ops-templates	<code>mlops-templates</code>	Required shared Azure ML pipeline helpers; this remains an upstream dependency.

For CV, NLP, Python SDK, Responsible AI, Terraform, or the explicit upstream opt-out, also import <https://github.com/Azure/mlops-project-template> as `mlops-project-template`. Do not substitute the standalone taxi repository for that nested template layout.

Visibility is configured separately for each GitHub repository. The personal accelerator is public at the time of this update; the taxi template remains private and requires an account with access to import it. Authenticate private imports in the Azure DevOps UI; never store an import token in source files. Imports are one-time copies, not automatic mirrors.

1. Navigate to [Azure DevOps](#) and the organization where you want to create the project. [Create a new organization](#) for your project, if needed.
2. Create a new project named `mlops-v2`.

Create new project



Project name *

mlops-v2



Description

Visibility



Public

Anyone on the internet can view the project. Certain features like TFVC are not supported.



Enterprise

[Members of your enterprise](#) can view the project.



Private

Only people you give access to will be able to view this project.

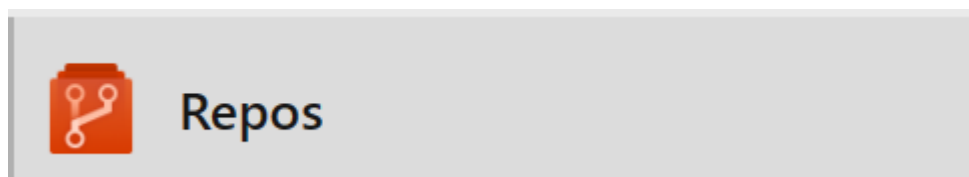


Advanced

Cancel

Create

3. Import the MLOps V2 repositories. In your new `mlops-v2` project, select the Repos section on the left side.



The default repo, `mlops-v2`, is empty. Under "Import a repository", select **Import**.

Import a repository

Import

Enter <https://github.com/alvinea28/own-ml-ops-v2> into the Clone URL field. Click import at the bottom of the page. This imports the personal accelerator into the Azure Repos repository named `mlops-v2` in this walkthrough.

Import a Git repository



Repository type

 Git 

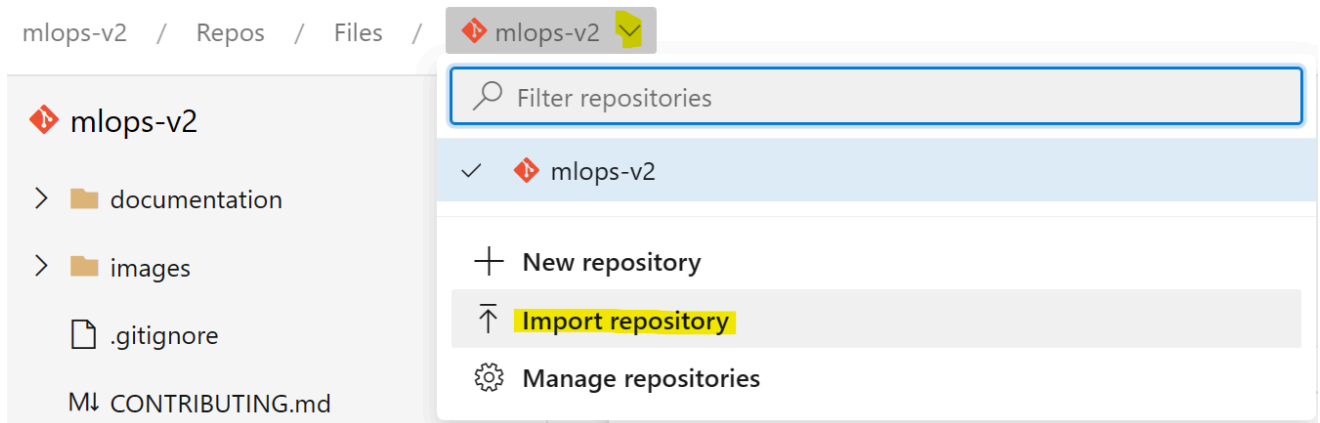
Clone URL *

<https://github.com/Azure/mlops-v2>

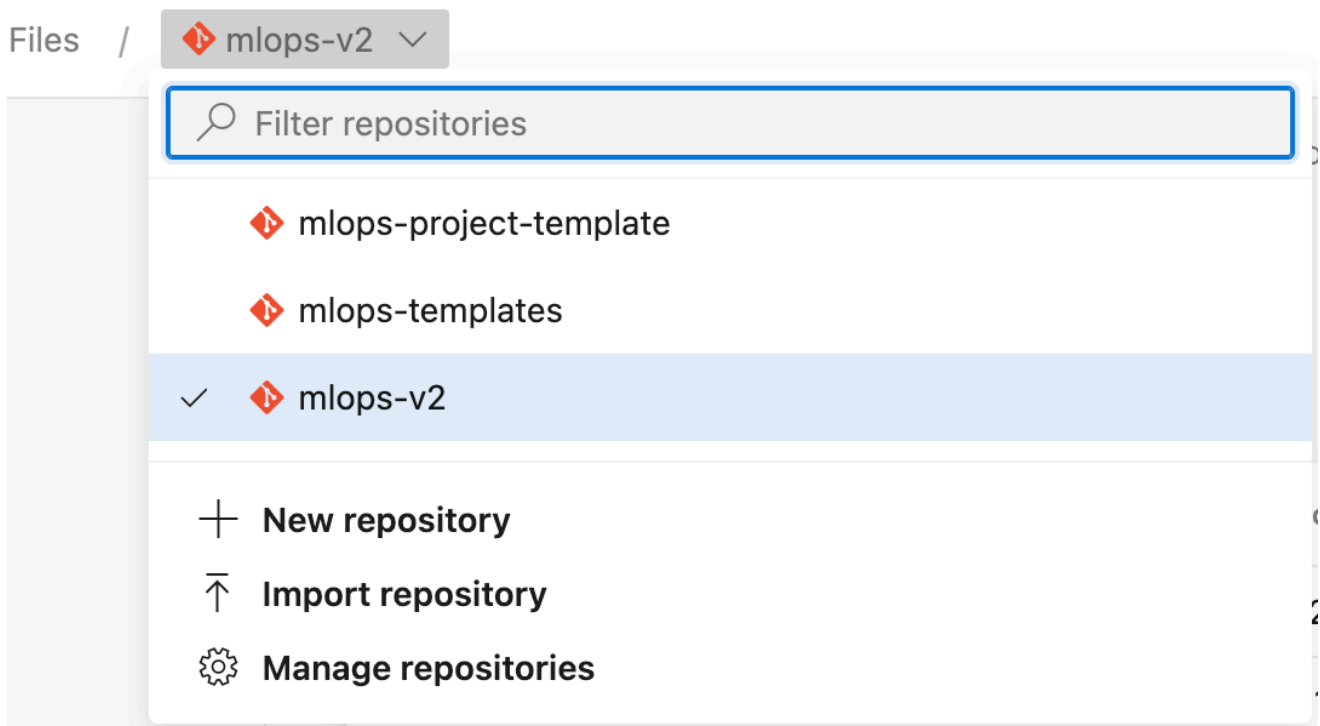
☐ Requires Authentication

At the top of the page, open the Repos drop-down and repeat the import for:

- <https://github.com/alvinea28/taxi-fare-regression>, naming its Azure Repos copy **taxi-fare-regression-template**. This is the reusable **source**, not the new application target.
- <https://github.com/Azure/mlops-templates>, naming its Azure Repos copy **mlops-templates**. This required helper repository has not been copied to the personal GitHub account.
- Optionally, <https://github.com/Azure/mlops-project-template> as **mlops-project-template** for the other scenarios described above. It has not been copied to the personal GitHub account and is not needed for the repaired taxi selection.



When done, you should see [all three MLOps V2 repositories](#) in your project.

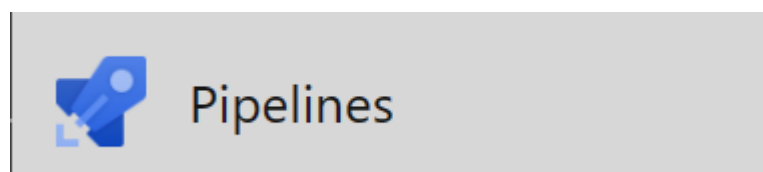


Important:

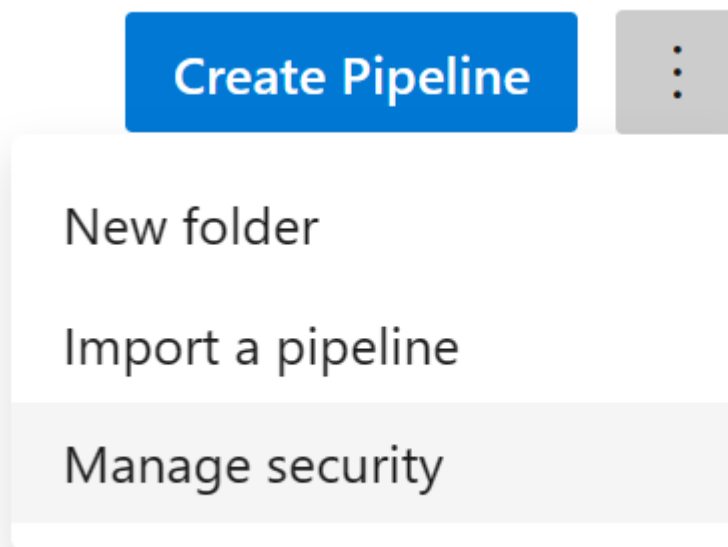
Azure DevOps may not import the three MLOps V2 repos with the default branch set to `main`. If not, select **Branches** under the **Repos** section on the left and [reset the default branch](#) to `main` for each of the three imported repos.

For an existing import, synchronize the latest personal `main` branch first. Selecting `main` in a saved Azure DevOps pipeline does not fetch later GitHub changes automatically.

4. Lastly, you will grant the MLOps Solution Accelerator permission to create new pipelines in the ML projects you will create. In your `mlops-v2` project, select the Pipelines section on the left side.



Select the three vertical dots next to **Create Pipeline** and select **Manage Security**.



Select the "<projectname> Build Service" account for your project under the Users section. Change the permissions for **Edit build pipeline** and **Create build pipeline** to **Allow**

Permissions for mlops-v2



Search for users or groups		mlops-v2 Build Service (REDACTED)	
▼ Azure DevOps Groups	BA Build Administrators	Administer build permissions	Not set ▼
	C Contributors	Delete build pipeline	Not set ▼
	PA Project Administrators	Delete builds	Not set ▼
	R Readers	Destroy builds	Not set ▼
	PA Project Collection Administrators	Edit build pipeline	Allow ▼ ✓
	PA Project Collection Build Administrators	Edit build quality	Allow ▼
	PA Project Collection Build Service Accounts	Manage build qualities	Not set ▼
	PA Project Collection Build Test Service Accounts	Manage build queue	Allow ▼
▼ Users		Override check-in validation by build	Allow ▼
	MS mlops-v2 Build Service (REDACTED) 🗑️	Queue builds	Not set ▼
		Retain indefinitely	Not set ▼
		Stop builds	Not set ▼
		Update build information	Allow ▼
		View build pipeline	Allow ▼
		View builds	Allow ▼

You are done cloning and configuring the MLOps V2 Solution Accelerator. Next, you will create a new ML project using the accelerator templates.

Create and Configure a New ML Project Repo

In this section, you will create your ML project repository, set permissions to allow the solution accelerator to interact with your project, and create service principals so your Azure pipelines can interact with Azure Machine Learning.

Creating the project repository

1. Open the **Repos** drop-down once more and select **New repository**. Create the empty **target** repository for your ML project; do not import a finished project into it. The initializer will populate its files from `taxi-fare-regression-template`. The existing walkthrough and screenshots call this target `taxi-fare-regression`; use `taxi-fare-regression-demo` or another new name if that name is already your working project.

Leave **Add a README** selected to initialize the repo with a `main` branch.

Use a new, empty application repository. If your working project is already named `taxi-fare-regression`, choose another name such as `taxi-fare-regression-demo` throughout the remaining steps. Never initialize over the working application or the `taxi-fare-regression-template` source repository.

Create a repository



Repository type

 Git 

Repository name *

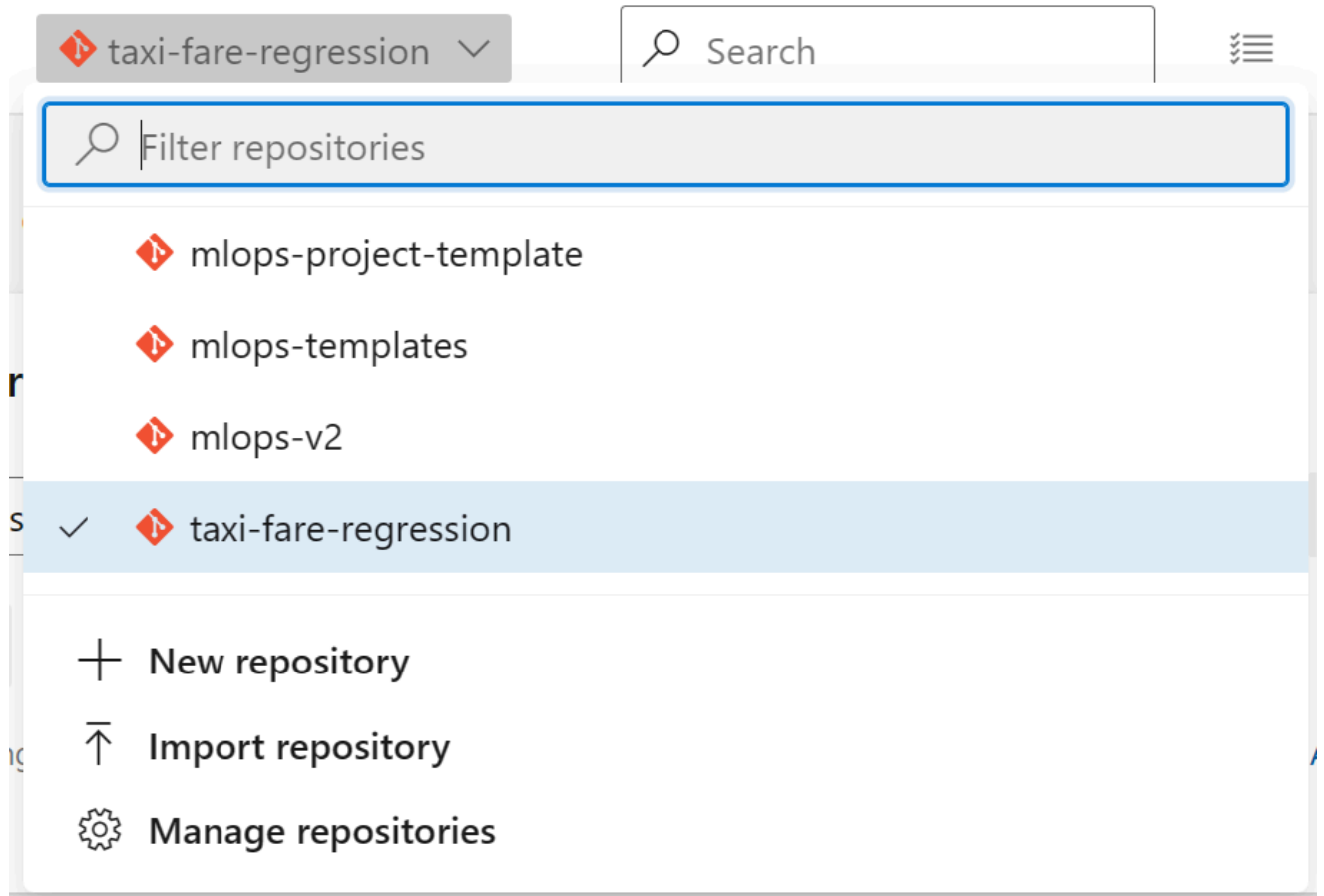
taxi-fare-regression

☒ Add a README

Add a .gitignore: None 

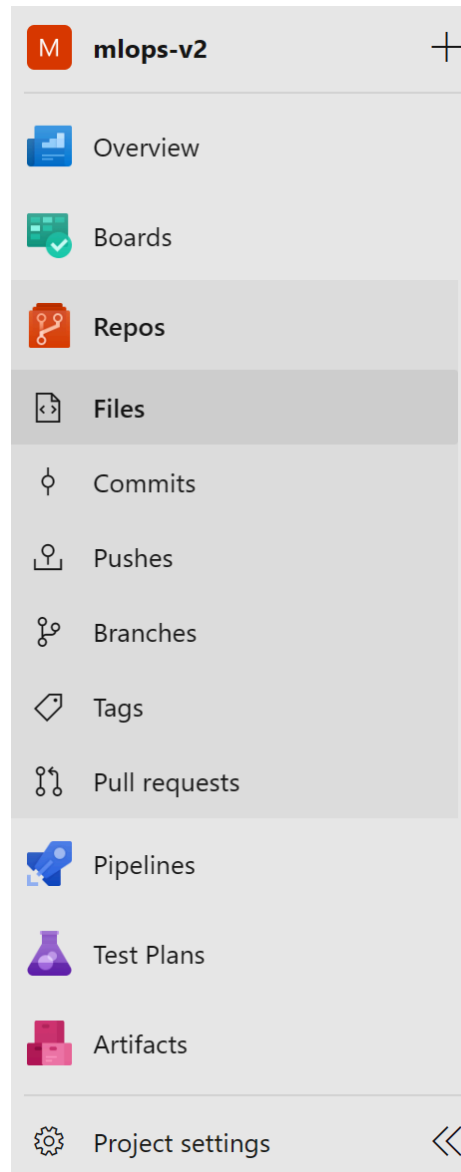
Your repository will be initialized with a  `main` branch.

You should now have your `taxi-fare-regression` repo and [all three MLOps V2 repositories](#) in your Azure DevOps project.



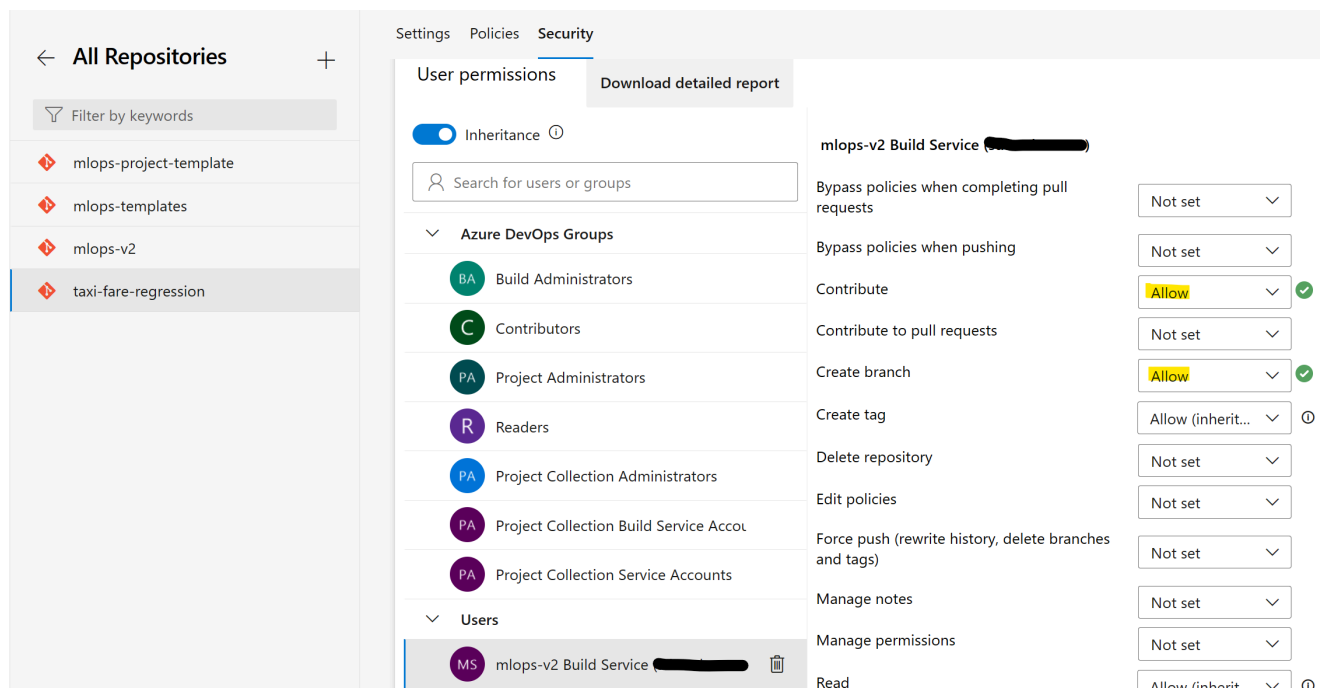
Setting project permissions

2. Next, set access permissions on your ML project repository. Open the **Project settings** at the bottom of the left hand navigation pane



Under the **Repos** section, select **Repositories**.

- Select the `taxi-fare-regression` repository.
- Select the **Security** tab.
- Under **User permissions**, select the "`<projectname> Build Service`" account for your project under the Users section.
- Change the permissions for **Contribute** and **Create branch** to **Allow**.



Initializing the new ML project repo

In this step, you will run an Azure DevOps pipeline, `initialise-project`, that will prompt you for the properties of the ML project you want to build including the ML scenario (classical, computer vision, or natural language processing), the interface you will use to interface with Azure ML (CLI or SDK), and the CI/CD tool and infrastructure provider your organization uses. When run, the pipeline will populate the empty repo you created in the previous steps with the correct elements of the template repos to build your project.

For the repaired taxi selection, the YAML checks out the imported source and your already-created target, copies the working source files, commits and pushes the generated project, and registers its pipelines. It does not create personal GitHub forks, create the target repository object, or deploy Azure resources during initialization.

3. Open the Pipelines section again and select **Create Pipeline** in the center of the page.

Azure DevOps **mlops-v2** / mlops-v2 / Pipelines

mlops-v2 +

- Overview
- Boards
- Repos
- Pipelines**
- Pipelines
- Environments
- Releases
- Library
- Task groups
- Deployment groups

Connect Select Configure Review

New pipeline

Where is your code?

- Azure Repos Git** (YAML)
Free private Git repositories, pull requests, and code search
- Bitbucket Cloud** (YAML)
Hosted by Atlassian
- GitHub** (YAML)
Home to the world's largest community of developers
- GitHub Enterprise Server** (YAML)
The self-hosted version of GitHub Enterprise
- Other Git**
Any generic Git repository
- Subversion**
Centralized version control by Apache

- Select **Azure Repos Git**
- Select the **mlops-v2** repository imported from **alvinea28/own-ml-ops-v2** (or **own-ml-ops-v2** if you kept that Azure Repos name)
- Select **Existing Azure Pipelines YAML file**
- Ensure the selected branch is **main**
- Select the `/.azuredevops/initialise-project.yml` file in the Path drop-down
- Click Continue

On the pipeline review page, drop-down the **Run** menu and select **Save** the pipeline before running it.

New pipeline

Review your pipeline YAML

✓ Connect ✓ Select ✓ Configure **Review**

mlops-v2 / .azuredevops/initialise-project.yml

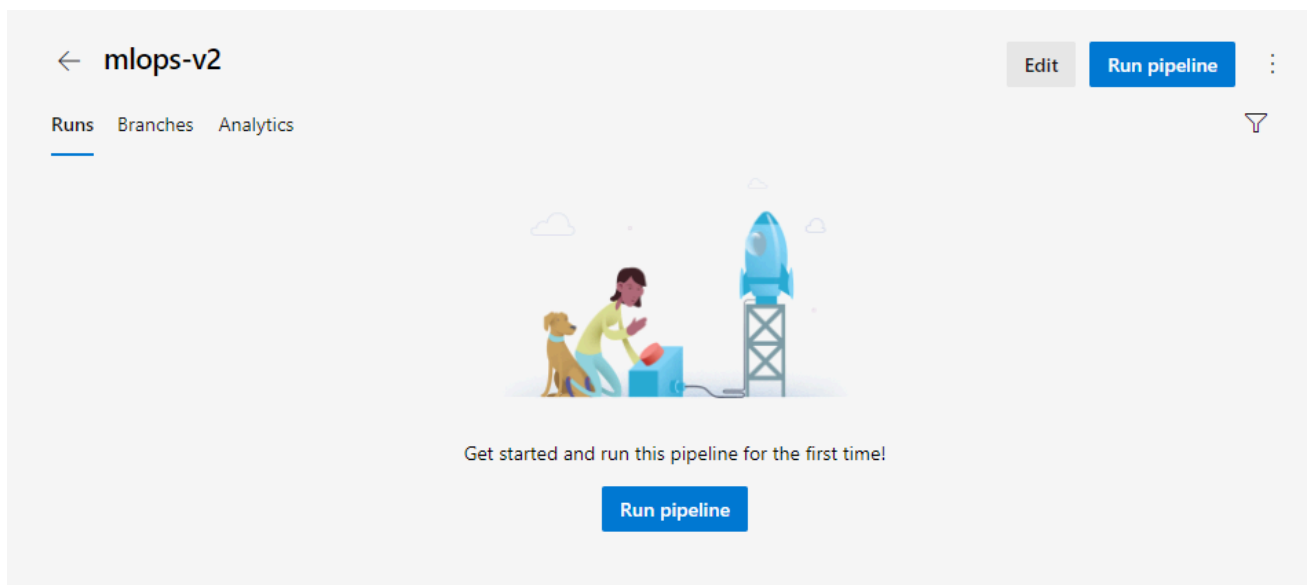
```

1 trigger: none
2
3 parameters:
4   - name: adoProjectName
5     displayName: Azure DevOps ProjectName
6     type: string
7

```

Variables Run Save Show assistant

Now select **Run pipeline**.



This action will run an Azure DevOps pipeline that prompts you for some parameters of your project. You can select the ML scenario, the interface the ML pipelines will use to interact with Azure Machine Learning, and the Infrastructure-as-Code provider for your organization. Below shows the parameter selection panel followed by explanations of each option:

Run pipeline

×

Select parameters below and manually run the pipeline

Branch/tag

▼

Select the branch, commit, or tag

Azure DevOps Project Name

Required

New Project Repository Name

Required

MLOps-V2 ADO Repository name

MLOps Project Template's repo

ML Project Type

☒ classical
☐ cv
☐ nlp

MLOps Interface

▼

Infrastructure Provider

☒ bicep
☐ terraform

Advanced options

Variables

This pipeline has no defined variables

>

Stages to run

Run as configured

>

Resources

Use latest version of all resources

>

☐ Enable system diagnostics

Cancel

Run

Project Parameters

- **Azure DevOps Project Name** : This is the name of the Azure DevOps project you are running the pipeline from. In this case, `mlops-v2` .
- **New Project Repository Name** (`repoName`): Your new empty **target** repository, for example **taxi-fare-regression-demo**. This must not be **taxi-fare-regression-template** or your existing working project.
- **Use repaired taxi template for classical / AML CLI v2 / Bicep** (`useRepairedTaxiTemplate`): Leave **true** for the repaired taxi walkthrough.
- **Repaired taxi template Azure Repos name** (`taxiTemplateRepoName`): Leave **taxi-fare-regression-template**, matching the import above.
- **Upstream template Azure Repos name (other scenarios)** (`mlopsProjectRepoName`): Leave **mlops-project-template**. This source is used only for other scenario selections or when the repaired taxi option is disabled; import it before choosing those paths.
- **ML Project type**:
 - Choose **classical** for a regression or classification project
 - Choose **cv** for a computer vision project
 - Choose **nlp** for natural language projects
- **MLOps Interface**: Select the interface to the Azure ML platform, either CLI or SDK.
 - Choose **aml-cli-v2** for the Azure ML CLI v2 interface. This is supported for all ML project types.
 - Choose **python-sdk-v2** to use the Azure ML python SDK v2 for training and deployment of your model. This is supported for Classical and NLP project types.
 - Choose **rai-aml-cli-v2** to use the Responsible AI cli tools for training and deployment of your model. This is supported only for Classical project types at this time.
- **Infrastructure Provider**: Choose the provider to use to deploy Azure infrastructure for your project.
 - Choose **Bicep** to deploy using Azure Bicep templates
 - Choose **terraform** to use terraform based templates.

For this repaired taxi walkthrough, select **classical**, **aml-cli-v2**, and **bicep** together. Other combinations continue to use the upstream project-template repository and do not inherit the standalone taxi fixes.

After selecting the parameters, click **Run** at the bottom of the panel. The first run of the pipeline will prompt you to grant access to the repositories you created.

 This pipeline needs permission to access 2 resources before this run can continue to Initialise Project View

Jobs
Permission needed

Name	Status	Duration
 Checkout repos, initialise your new one and create pipelines	Waiting	

Sources

Repository	Branch / tag	Version	Related
mlops-v2 Azure Repos	 main	663ca9cf	None
mlops-project-template Azure Repos	 main	2915b1ed	
taxi-fare-regression Azure Repos	 main	11c0acf5	

Click **View** to see the permissions waiting for review.

Waiting for review



Initialise Project

**Permission**
Permission needed

 taxi-fare-regression
Repository

Permit

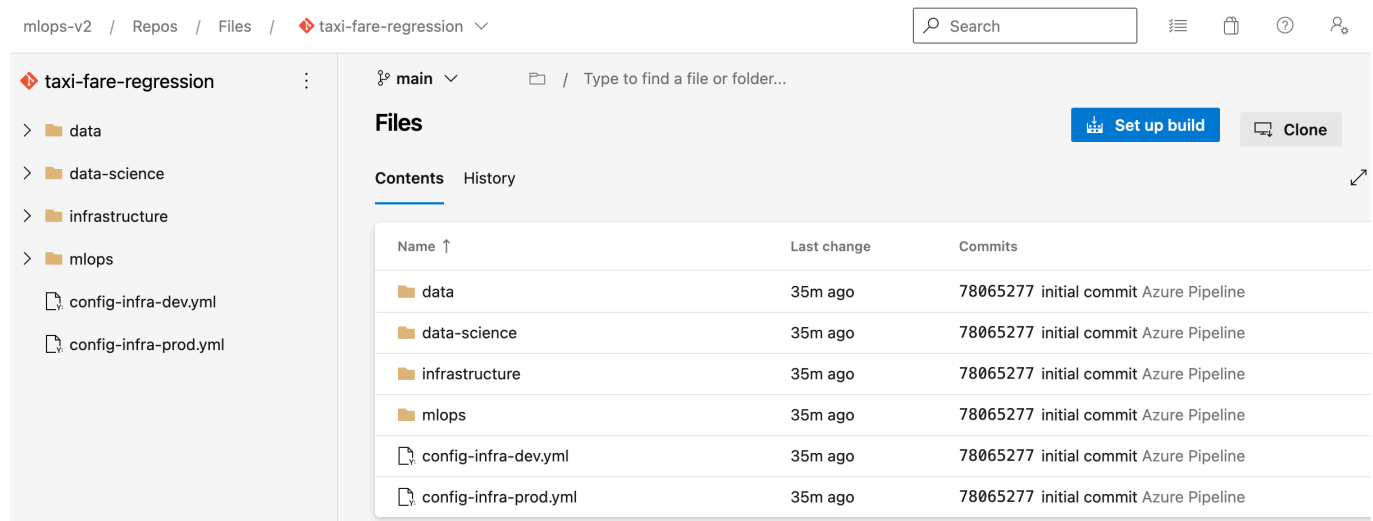
**Permission**
Permission needed

 mlops-project-tem...
Repository

Permit

For each of the repos, click **Permit** waiting for review.

The pipeline run should take a few minutes. When the pipeline run is complete and successful, go back to **Repos** and look at the contents of your ML project repo, `taxi-fare-regression`. The solution accelerator has populated the project repository according to your configuration selections.



The structure of the project repo is as follows:

File	Purpose
/data	Sample data for the example project
/data-science	Contains python code for the data science workflow
/infrastructure	IaC code for deploying the Azure Machine Learning infrastructure
/mlops	Azure DevOps pipelines and Azure Machine Learning pipelines for orchestrating deployment of infrastructure and ML workflows.
config-infra-dev.yml	Configuration file to define dev environment resources
config-infra-prod.yml	Configuration file to define production environment resources

Create and Configure Service Principals and Connections

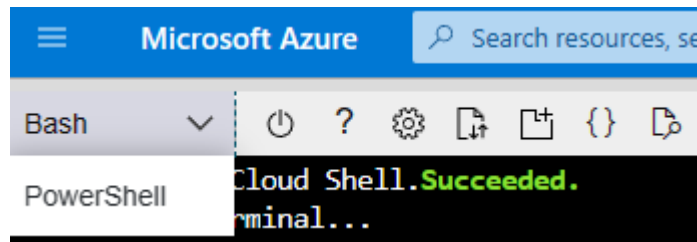
For Azure DevOps pipelines to create Azure Machine Learning infrastructure and deploy and execute Azure ML pipelines, it is necessary to create an Azure service principal for each Azure ML environment (Dev and/or Prod) and configure Azure DevOps service connections using those service principals.

Security Best Practice: Azure DevOps now supports **workload identity federation** as a more secure alternative to service principal secrets. Workload identity federation uses OpenID Connect (OIDC) to establish trust without storing long-lived secrets. For production deployments, consider using workload identity federation instead of the service principal methods described below. See [Microsoft's documentation](#) for setup instructions.

These service principals can be created using one of the two methods below:

Create Service Principal from Azure Cloud Shell

1. Launch the [Azure Cloud Shell](#) . (If this the first time you have launched the cloud shell, you will be required to create a storage account for the cloud shell.)
2. If prompted, choose **Bash** as the environment used in the Cloud Shell. You can also change environments in the drop-down on the top navigation bar



3. Copy the bash commands below to your computer and update the **projectName**, **subscriptionId**, and **environment** variables with the values for your project. If you are creating both a Dev and Prod environment you will need to run this script once for each environment, creating a service principal for each. This command will also grant the **Contributor** role to the service principal in the subscription provided. This is required for Azure DevOps to properly deploy resources to that subscription.

```
projectName="<your project name>"
roleName="Contributor"
subscriptionId="<subscription Id>"
environment="<Dev|Prod>" #First letter should be capitalized
servicePrincipalName="Azure-ARM-${environment}-${projectName}"
# Verify the ID of the active subscription
echo "Using subscription ID $subscriptionId"
echo "Creating SP for RBAC with name $servicePrincipalName, with role $roleName and in scopes
/subscriptions/$subscriptionId"
az ad sp create-for-rbac --name $servicePrincipalName --role $roleName --scopes
/subscriptions/$subscriptionId
echo "Please ensure that the information created here is properly save for future use."
```

4. Copy your edited commmands into the Azure Shell and run them (**Ctrl** + **Shift** + **v**).

```
kevin@Azure:~$ projectName="Sample_Project_Name"
roleName="Contributor"
subscriptionId="fbf19cb9-0e29-493b-abed-82713041d5a3"
environment="Dev" #First letter should be capitalized
servicePrincipalName="Azure-ARM-${environment}-${projectName}"
# Verify the ID of the active subscription
echo "Using subscription ID $subscriptionID"
echo "Creating SP for RBAC with name $servicePrincipalName, with role $roleName and in scopes /subscriptions/$subscriptionId"
az ad sp create-for-rbac --name $servicePrincipalName --role $roleName --scopes /subscriptions/$subscriptionId
echo "Please ensure that the information created here is properly save for future use."
Using subscription ID
Creating SP for RBAC with name Azure-ARM-Dev-Sample_Project_Name, with role Contributor and in scopes /subscriptions/fbf19cb9-0e29-493b-abed-82713041d5a3
Creating 'Contributor' role assignment under scope '/subscriptions/fbf19cb9-0e29-493b-abed-82713041d5a3'
The output includes credentials that you must protect. Be sure that you do not include these credentials in your code or check the credentials into your source control.
For more information, see https://aka.ms/azadsp-cli
{
  "appId": "074da5fb-8ffc-42cd-af0f-b48f1d1cb9e8",
  "displayName": "Azure-ARM-Dev-Sample_Project_Name",
  "password": "6QK8Q<CKo3x1jQcyrTQA3pC31IKImIznde9cLaau",
  "tenant": "20a1d97b-1261-47c9-8cec-adba8291275f"
}
Please ensure that the information created here is properly save for future use.
kevin@Azure:~$
```

5. After running these commands you will be presented with information related to the service principal. Save this information to a safe location, it will be used later in the demo to configure Azure DevOps.

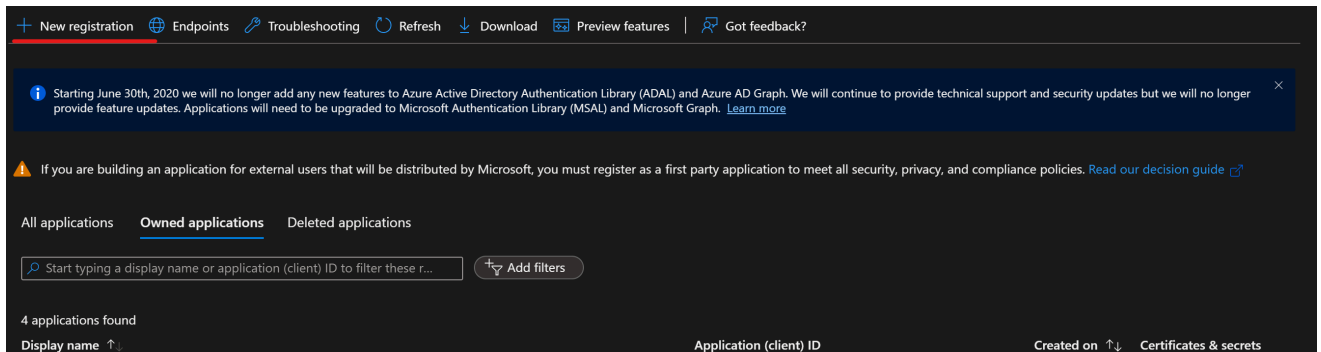
```
{
  "appId": "<application id>",
  "displayName": "Azure-ARM-dev-Sample_Project_Name",
```

```
"password": "<password>",
"tenant": "<tenant id>"
}
```

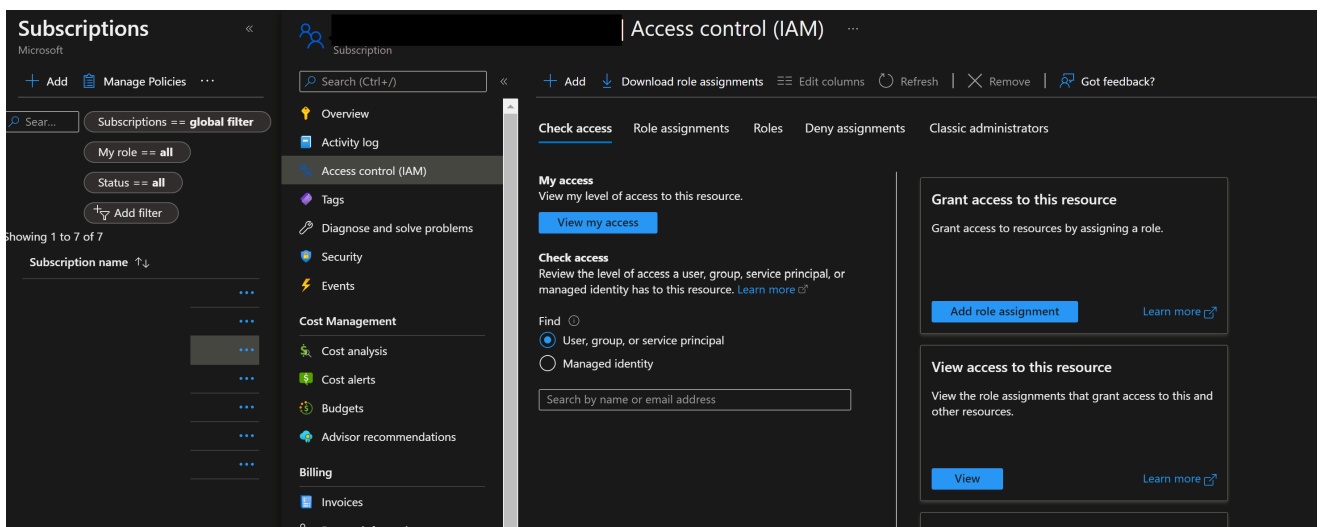
- Repeat **step 3** if you are creating service principals for Dev and Prod environments.
- Close the Cloud Shell once the service principals are created.

Create Service Principal from the Azure Portal

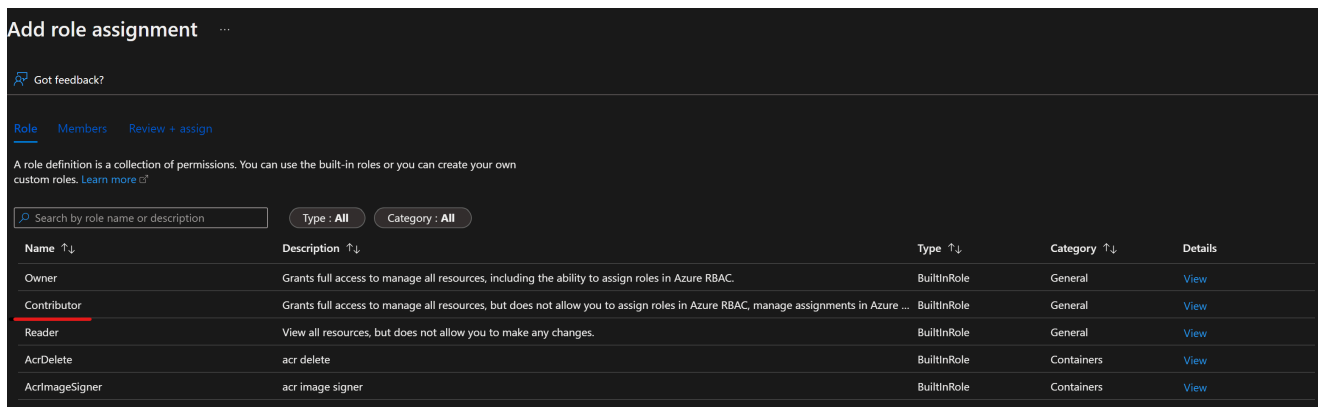
- Navigate to [Azure App Registrations](#)
- Select **New registration**.



- Go through the process of creating a Service Principle (SP) selecting "Accounts in any organizational directory (Any Azure AD directory - Multitenant)" and name it "Azure-ARM-Dev-ProjectName". Once created, repeat and create a new SP named "Azure-ARM-Prod-ProjectName". Please replace "ProjectName" with the name of your project so that the service principal can be uniquely identified.
- Go to "Certificates & Secrets" and add for each SP "New client secret", then store the value and secret separately.
- To assign the necessary permissions to these principals, select your respective [subscription](#) and go to **Access control (IAM)**. Select +Add then select "Add Role Assignment".



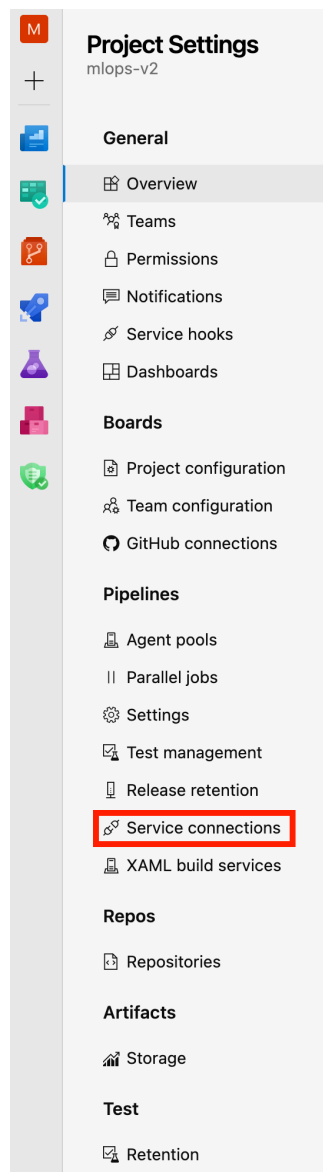
- Select Contributor and add members selecting + Select Members. Add the member "Azure-ARM-Dev-ProjectName" as create before.



7. Repeat step here, if you deploy Dev and Prod into the same subscription, otherwise change to the prod subscription and repeat with "Azure-ARM-Prod-ProjectName". The basic SP setup is successfully finished.

Create Service Connections

Select **Project Settings** at the bottom left of the project page and select **Service connections**.



Select **Create service connection**

- For service, select **Azure Resource Manager** and **Next**
- For identity type, select **App registration (automatic)**
- For Credential, select **Workload identity Federation**
- for Subscription, select your subscription from the drop-down and select **Save**

New Azure service connection ✕

Azure Resource Manager

Identity type

App registration (automatic)
Recommended

[Need help choosing a connection type?](#)

Credential

Workload identity federation
Recommended

Scope level

☒ Subscription
☐ Management Group
☐ Machine Learning Workspace

Subscription

If you can't find subscription you need, please [create manually](#).

Resource group

Specify to limit access to the chosen resource group only.

Service Connection Name

Azure-ARM-Prod

Service Management Reference (optional)

Provide a Service Management Reference if your organization requires one to be configured on Entra ID App registrations.

Description (optional)

Security

☐ Grant access permission to all pipelines

[Learn more](#)
[Troubleshoot](#)

Back
Save

Complete the new service connection configuration using the information from your tenant, subscription, and the service principal you created for Prod.

Name this service connection **Azure-ARM-Prod**. Check **Grant access permission to all pipelines**. and click **Verify and save**.

Create Azure DevOps Environment

The pipelines in each branch of your ML project repository will depend on an Azure DevOps environment. These environments should be created before deployment.

To create the prod environment, select **Pipeline** in the left menu and **Environments**. Select **New environment**

New environment



Name ⓘ

prod

Description ⓘ

Describe the environment

Resource



None

You can add resources later



Kubernetes

Add Kubernetes namespace



Virtual machines

Manage virtual machines

Name the new environment `prod` and click **Create**. The environment will initially be empty and indicate "Never deployed" but this status will update after the first deployment.

The configuration of your new ML project repo is complete and you are ready to deploy your Azure Machine Learning infrastructure and deploy ML training and model deployment pipelines in the next section.

Deploy and Execute Azure Machine Learning Pipelines

Now that your ML project is created, this last section will guide you through executing Azure DevOps pipelines created for you that will first deploy Azure Machine Learning infrastructure for your project then deploy a model training pipeline followed by a model deployment pipeline.

Each pipeline may have different roles associated with its deployment and management. For example, infrastructure by your IT team, model training by your data scientists and ML engineers, and model deployment by ML engineers. Likewise, depending on the environments and project branches you have created, you may deploy infrastructure for both **dev** and **prod** Azure ML infrastructure, with data scientists developing the training pipeline in the **dev** environment and branch and, when the model is acceptable, opening a pull request to the **main** branch to merge updates and run the model deployment pipeline in the **prod** environment.

Depending on the options you chose when initializing the project, you should have one infrastructure deployment pipeline, one model training pipeline, and one or two model deployment pipelines in your ML project. Model deployment options are online-endpoint for near real-time scoring and batch-endpoint for batch scoring. To see all pipelines in your project, select the **Pipelines** section from the left navigation menu, then **Pipelines**, then the **All** tab. For the example project in this guide, you should see:

Pipelines New pipeline

Recent **All** Runs Filter pipelines

All pipelines New folder

Name
taxi-fare-regression
infrastructure bicep-ado-deploy-infra No runs yet
mlops deploy-batch-endpoint-pipeline No runs yet deploy-model-training-pipeline No runs yet deploy-online-endpoint-pipeline No runs yet
mlops-v2 #20230331.1 • adding context/clarification for when using WSL (#89)

Deploy Azure Machine Learning Infrastructure

The first task for your ML project is to deploy Azure Machine Learning infrastructure in which to develop your ML code, define your datasets, define your ML pipelines, train models, and deploy your models in production. This pipeline deployment is typically managed by your IT group responsible for ensuring that the subscription is able to create the infrastructure needed. The infrastructure is created by executing the Azure DevOps deploy-infra pipeline. Before doing this, you will customize environment files that define unique Azure resource groups and Azure ML workspaces for your project.

To do this, go back to **Repos** and your ML project repo, in this example, `taxi-fare-regression`. You will see two files in the root directory, `config-infra-prod.yml` and `config-infra-dev.yml`.

ml-ops-v2 / Repos / Files / taxi-fare-regression

Search

taxi-fare-regression

- data
- data-science
- infrastructure
- mlops
- config-infra-dev.yml
- config-infra-prod.yml

main

Type to find a file or folder...

Files

Contents History

Name ↑	Last change	Commits
data	35m ago	78065277 initial commit Azure Pipeline
data-science	35m ago	78065277 initial commit Azure Pipeline
infrastructure	35m ago	78065277 initial commit Azure Pipeline
mlops	35m ago	78065277 initial commit Azure Pipeline
config-infra-dev.yml	35m ago	78065277 initial commit Azure Pipeline
config-infra-prod.yml	35m ago	78065277 initial commit Azure Pipeline

Set up build Clone

Important:

Note that `config-infra-prod.yml` and `config-infra-dev.yml` files use default region as **eastus** to deploy resource group and Azure ML Workspace. If you are using Free/Trial or similar learning purpose subscriptions, you must do one of the below -

1. If you decide to use **eastus** region, ensure that your subscription(s) have a quota/limit of up to 64 vCPUs for **Standard DSv3 Family vCPUs**. The default compute cluster uses **STANDARD_D16S_V3** (16 vCPUs per node, up to 4 nodes = 64 vCPUs max). Visit Subscription page in Azure Portal as shown below to validate this.

Visual Studio Premium with MSDN | Usage + quotas

Search

Budgets

Advisor recommendations

Billing

- Invoices
- External services
- Payment methods
- Partner information

Settings

- Programmatic deployment
- Resource groups
- Resources
- Preview features
- Usage + quotas

New Quota Request Refresh Download

You can now set up alerts for your Quota usage and receive notifications. Simply click on any Quota to create one. [Learn More.](#)

Recommended To view and manage quotas across all your subscriptions from a central location, go to [Azure Quotas.](#)

dsv2 Provider : Compute Region : East US Usage : Show all

Showing 1 to 3 of 3 records in 1 groups.

Group by usage

Quota name	Region	Subscription	Current Usage	Adjustable
☐ Standard DSv2 Family vCPUs	East US	Visual Studio Premium with	0% 0 of 20	Yes
☐ Standard DSv2 Promo Family vCPUs	East US	Visual Studio Premium with	0% 0 of 20	No
☐ Standard NDsv2 Fa... (troubleshoot)	East US	Visual Studio Premium with	0% 0 of 0	Yes

2. If not, you should change it to a region where **Standard DSv3 Family vCPUs** has sufficient quota.
3. You can easily change the VM SKU by editing the `aml_compute_sku` parameter in your config file:

- `config-infra-prod.yml` or `config-infra-dev.yml` - set `aml_compute_sku: <YOUR_SKU>` (e.g., `STANDARD_D4S_V3`)
- This works for both Bicep and Terraform deployments

4. For ML pipeline compute (separate from infrastructure), you may need to edit:

- `mlops-templates/aml-cli-v2/mlops/devops-pipelines/deploy-model-training-pipeline.yml` - for ML pipeline compute

- `mlops-project-template/classical/aml-cli-v2/mlops/devops-pipelines/deploy-batch-endpoint-pipeline.yml`
- `/mlops-project-template/classical/aml-cli-v2/mlops/azureml/deploy/online/online-deployment.yml`

Note: The default infrastructure SKU is **STANDARD_D16S_V3** (3rd generation). ML pipelines may use different SKUs like **Standard_D4s_v5**. Adjust based on your quota and requirements.

Making sure you are in the **main** branch, click on `config-infra-prod.yml` to open it.

Under the Global section, you will see properties for `namespace`, `postfix`, and `location`.

```
# Prod environment
variables:
  # Global
  ap_vm_image: ubuntu-20.04

  namespace: mlopsv2 #Note: A namespace with many characters will cause storage account creation to fail due
to storage account names having a limit of 24 characters.
  postfix: 0001
  location: eastus
  environment: prod
  enable_aml_computecluster: true
```

The two properties `namespace` and `postfix` will be used to construct a unique name for your Azure resource group, Azure ML workspace, and associated resources. The naming convention for your resource group will be `rg-<namespace>-<postfix>prod`. The name of the Azure ML workspace will be `mlw-<namespace>-<postfix>prod`. The `location` property will be the Azure region into which to provision these resources.

Edit `config-infra-prod.yml` to set the variables for your environment. You can clone the repo, edit the file, and push/PR to make the change or select **Edit** in the upper right of the screen to edit the file within Azure DevOps. If editing in place, change `namespace`, `postfix`, and `location` to your preferences and click **Commit**

If the `enable_aml_computecluster` property is set to true, the infra deployment pipeline will pre-create Azure ML compute clusters for your training. In the case of CV or NLP scenarios, it will create both CPU-based and GPU-based compute clusters so ensure that your subscription has GPU compute available.

Now you are ready to run the infrastructure deployment pipeline. Open the **Pipelines** section again and select **New pipeline** in the upper right of the page.

mlops-v2

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Connect

Select

Configure

Review

New pipeline

Where is your code?

Azure Repos Git

YAML

Free private Git repositories, pull requests, and code search

Bitbucket Cloud

YAML

Hosted by Atlassian

GitHub

YAML

Home to the world's largest community of developers

GitHub Enterprise Server

YAML

The self-hosted version of GitHub Enterprise

Other Git

Any generic Git repository

Subversion

Centralized version control by Apache

- Select **Azure Repos Git**
- Select the **taxi-fare-regression** repository
- Select **Existing Azure Pipelines YAML file**
- Ensure the selected branch is **main**
- Select the `/infrastructure/pipelines/bicep-ado-deploy-infra.yml` file in the Path drop-down
- Click Continue

Now you will see the pipeline details.

✓ Connect

✓ Select

✓ Configure

Review

New pipeline

Review your pipeline YAML

Variables

Run

taxi-fare-regression / infrastructure/pipelines/bicep-ado-deploy-infra.yml

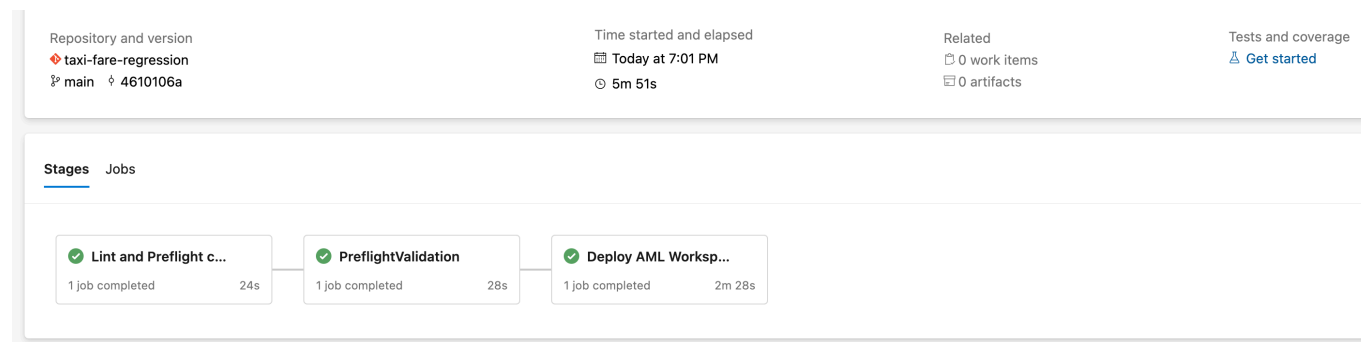
Show assistant

```

1  Copyright (c) Microsoft Corporation. All rights reserved.
2  # Licensed under the MIT License.
3
4  variables:
5  - ${{ if eq(variables['Build.SourceBranchName'], 'main') }}:
6  |   # 'main' branch: PRD environment
7  |   template: ../../config-infra-prod.yml
8  - ${{ if ne(variables['Build.SourceBranchName'], 'main') }}:
9  |   # 'develop' or feature branches: DEV environment
10 |   template: ../../config-infra-dev.yml
11

```

Click **Run** to execute the pipeline. This will take a few minutes to finish. When complete, you can view the pipeline jobs and tasks by selecting **Pipelines** then **taxi-fare-regression** under **Recently run pipelines**.



The pipeline should create the following artifacts which you can view in your Azure subscription:

- Resource Group for your Workspace
- Azure Machine Learning Workspace and associated resources including Storage Account, Container Registry, Application Insights, and Keyvault
- Inside the workspace, an AmlCompute cluster will be created

Your Azure Machine Learning infrastructure is now deployed and you are ready to deploy an ML model training pipeline.

Deploy Azure Machine Learning Model Training Pipeline

The solution accelerator includes code and data for a sample end-to-end machine learning pipeline which trains a linear regression model to predict taxi fares in NYC. The pipeline is made up of multiple steps for data prep, training, model evaluation, and model registration. Sample pipelines and workflows for the Computer Vision and NLP scenarios will have different steps.

In this section you will execute an Azure DevOps pipeline that will create and run an Azure Machine Learning pipeline. Together, they perform the following steps:

- Connect to the Azure Machine Learning workspace created by the infrastructure deployment
- Create a compute cluster for training in the workspace (refer to section below to create compute instances with or without managed identity)
- Register the training dataset in the workspace
- Prepare data for training
- Registers a custom python environment with the packages required for this model
- Train a linear regression model to predict taxi fares
- Evaluate the model on the test dataset against the performance of any previously-registered models
- If the new model performs better, register the model as an MLflow model in the workspace for later deployment

Create Compute Instances with SystemAssigned/UserAssigned/No Managed Identity

In order to create a compute instance with or without managed identity, you can leverage the `/mlops-templates/templates/python-sdk-v2/create-compute-instance.yml` located within the **mlops-templates** repository.

If you want to create a **compute instance without a managed identity** reference, you can add the following snippet with your own parameters to the `/mlops/devops-pipelines/deploy-model-training-pipeline.yml` pipeline definition:

```
- template: templates/python-sdk-v2/create-compute-instance.yml@mlops-templates
  parameters:
    instance_name: compute-instance-a
    size: Standard_D4s_v5
    location: canadacentral
    description: compute instance a
```

In order to **create a system-assigned managed identity** and assign it your compute instance during creation, the above snippet can be adjusted as follows:

```
- template: templates/python-sdk-v2/create-compute-instance.yml@mlops-templates
  parameters:
    instance_name: compute-instance-a
    size: Standard_D4s_v5
    location: canadacentral
    description: compute instance a
    identity_type: SystemAssigned
```

Lastly, to leverage a **user-assigned managed identity** for your compute, the following snippet can be used and adjusted as needed:

```
- template: templates/python-sdk-v2/create-compute-instance.yml@mlops-templates
  parameters:
    instance_name: compute-instance-a
    size: Standard_D4s_v5
    location: canadacentral
    description: compute instance a
    identity_type: UserAssigned
    user_assigned_identity: e12c9326-0618-4036-a0a7-ad3bb396dc97
```

To deploy the model training pipeline, open the **Pipelines** section again and select **New pipeline** in the upper right of the page

- Select **Azure Repos Git**
- Select the **taxi-fare-regression** repository
- Select **Existing Azure Pipelines YAML file**
- Ensure the selected branch is **main**
- Select the `/mlops/devops-pipelines/deploy-model-training-pipeline.yml` file in the Path drop-down
- Click Continue

Next you can see the pipeline details.

✓ Connect ✓ Select ✓ Configure **Review**

New pipeline

Review your pipeline YAML

Variables **Run** ▾

taxi-fare-regression / mlops/devops-pipelines/deploy-model-training-pipeline.yml

```

1  | Copyright (c) Microsoft Corporation. All rights reserved.
2  | # Licensed under the MIT License.
3  |
4  | name: deploy-model-training-pipeline
5  |
6  | > variables: --
15 |
16 |   trigger: none
17 |
18 | > pool: --
20 |
21 | > resources: --
27 |
28 |   stages:
29 |     - stage: DeployTrainingPipeline
30 |       displayName: Deploy Training Pipeline
31 |       jobs:
32 |         - job: DeployTrainingPipeline

```

Click **Run** to execute the pipeline. This will take several minutes to finish.

Important:

- It is noted that, with Free/Trial and other learning purpose subscriptions pipeline execution might take up to 90 minutes.
- In case, your model training pipeline execution in DevOps fails after running for a long period of time, but you have validated that a model is registered in Model Registry, it confirms that your model training is Azure ML was successful.
- This issue with longer running pipelines in Azure DevOps has been reported and is being tracked by product teams.

When complete, you can view the pipeline jobs and tasks by selecting **Pipelines** then **taxi-fare-regression** under **Recently run pipelines**. The pipeline run will be tagged `#deploy-model-training-pipeline`. Drill down into the pipeline run to see the **DeployTrainingPipeline** job. Click on the job to see pipeline run details.

Jobs			
Name	Status	Duration	
✓ DeployTrainingPipeline	Success	🕒 19m 9s	

Sources			
Repository	Branch / tag	Version	Related
taxi-fare-regression Azure Repos	📁 main	86571b9a	None
mlops-templates Azure Repos	📁 main	aa35e600	

Now you can open your Azure Machine Learning workspace to see the training run artifacts. Open a browser to <https://ml.azure.com> and login with your Azure account. You should see your Azure Machine Learning workspace under the **Workspaces** tab on the left. Your workspace name will have a name built from the options you chose in the `config-infra-prod.yml` file with the format **mlw-(namespace)-(postfix)(environment)**. For example, **mlw-mlopsv2-0001prod**. Click on your workspace. You will be presented with the Azure ML Studio home page for the workspace showing your training pipeline jobs under **Recent jobs**.

Recent jobs ...								View all
Display name	☆	Experiment	Status	Created on	Duration	Created by	Job type	
register_model	⋮	prod_taxi_fare_train_main	✓ Completed	Apr 5, 2023 2:08 PM	23s	Service Principal	Pipeline s...	
evaluate_model	⋮	prod_taxi_fare_train_main	✓ Completed	Apr 5, 2023 2:07 PM	21s	Service Principal	Pipeline s...	
train_model	⋮	prod_taxi_fare_train_main	✓ Completed	Apr 5, 2023 2:06 PM	43s	Service Principal	Pipeline s...	
prep_data	⋮	prod_taxi_fare_train_main	✓ Completed	Apr 5, 2023 1:53 PM	51s	Service Principal	Pipeline s...	
prod_taxi_fare_run_117	⋮	prod_taxi_fare_train_main	✓ Completed	Apr 5, 2023 1:53 PM	15m 14s	Service Principal	Pipeline	

You can now explore the artifacts of the pipeline run from the workspace navigation menus on the left:

- Select **Data** to see and explore the registered `taxi-data` Data asset.
- Select **Jobs** to see and explore the `prod_taxi_fare_train_main` Experiment.
- Select **Pipelines** to see and explore the `prod_taxi_fare_run` pipeline run.
- Select **Environments** then **Custom environments** to see the custom `taxi-train-env` environment registered by the pipeline.
- Select **Models** to see the `taxi-model` registered by the training pipeline.
- Select **Compute** then **Compute clusters** to see the `cpu-cluster` created by the training pipeline.

This section demonstrated end-to-end model training in Azure DevOps/Azure ML pipelines, creating all necessary assets as part of the pipeline. With a trained model registered in the Azure Machine Learning workspace, the next section will guide you through deploying the model to as either a real-time endpoint or batch scoring endpoint.

Deploy Azure Machine Learning Model Deployment Pipeline

In this section you will execute an Azure DevOps pipeline that will create and run an Azure Machine Learning pipeline that deploys your trained model to an endpoint. This can be a online managed (real-time) endpoint called by an application to score new data or a batch managed endpoint to score larger blocks of new data. In this example, there are two sample model deployment pipelines provided, one for online endpoint and one for batch endpoint. You can deploy one or the other or both.

For each type of endpoint, the deployment steps are essentially the same:

- Connect to the Azure Machine Learning workspace created by the infrastructure deployment
- Create a new compute cluster (batch managed endpoint only)
- Create an endpoint in Azure Machine Learning for the model deployment
- Create a deployment of the trained model on the new endpoint
- Update traffic allocation for the endpoint
- Test the deployment with sample data

For batch managed endpoint deployments, a new compute cluster is created to process batch scoring requests. For online managed endpoints, the compute to process requests is managed by Azure Machine Learning.

To deploy the model deployment pipeline, open the **Pipelines** section again and select **New pipeline** in the upper right of the page

- Select **Azure Repos Git**
- Select the **taxi-fare-regression** repository

- Select **Existing Azure Pipelines YAML file**
- Ensure the selected branch is **main**
- Select `/mlops/devops-pipelines/deploy-online-endpoint-pipeline.yml` or `/mlops/devops-pipelines/deploy-batch-endpoint-pipeline.yml` in the Path drop-down depending on your choice
- Click Continue

Again, from the pipeline details path, click **Run** to execute the pipeline. This will take several minutes to finish. When complete, you can view the pipeline jobs and tasks by selecting **Pipelines** then **taxi-fare-regression** under **Recently run pipelines**. The pipeline run will be tagged `#deploy-online-endpoint-pipeline` or `#deploy-batch-endpoint-pipeline`. Drill down into the pipeline run to see the **DeployOnlineEndpoint** or **DeployBatchEndpoint** job and click on the job to see pipeline run details.

Once the deployment pipeline execution is complete, open your Azure Machine Learning workspace to see the deployed endpoints. Select **Endpoints** from the workspace navigation menu on the left. By default, you will see a list of deployed **Online endpoints**. If you chose to deploy the sample online endpoint pipeline, you should see your `taxi-online-(namespace)(postfix)prod` endpoint.

Endpoints

Real-time endpoints Batch endpoints

+ Create ↻ Refresh 🗑 Delete 📄 Edit columns ↺ Reset view

Showing 1-1 endpoints

Name	☆	Description	Created on	Created by	Updated on ↓	Compute type
taxi-online-sdtest0042prod		taxi cost online endpoint	Apr 5, 2023 7:16 PM	Service Principal	Apr 5, 2023 7:27 PM	Managed

Click on this endpoint instance to explore the details of the online endpoint model deployment.

If you deployed the batch managed endpoint, select **Batch endpoints** on the **Endpoints** page to see your `taxi-batch-(namespace)(postfix)prod` endpoint. Click on this endpoint instance to explore the details of the batch endpoint model deployment.

Endpoints

Real-time endpoints **Batch endpoints**

+ Create ↻ Refresh 🗑 Delete 📄 Edit columns ↺ Reset view

Showing 1-1 endpoints

Name	Description	Created on	Created by	Updated on	Tags
taxi-batch-sdtest0042prod	taxi cost batch endpoint	Apr 5, 2023 7:44 PM	Service Principal	Apr 5, 2023 7:44 PM	

For the batch endpoint, you can also select **Compute** and ***Compute clusters** to see the cluster created to support batch request processing.

Compute

[Compute instances](#)
[Compute clusters](#)
[Kubernetes clusters](#)
[Attached computes](#)

[+ New](#)
[Refresh](#)
[Delete](#)
[Edit columns](#)
[Reset view](#)
[View quota](#)

Name	☆	State	Size
batch-cluster		✓ Succeeded (0 nodes)	STANDARD_DS3_V2
cpu-cluster		✓ Succeeded (0 nodes)	STANDARD_DS3_V2

This section demonstrated use of a pipeline to deploy a trained model to a managed online or managed batch endpoint in Azure Machine Learning.

The single-environment deployment of this MLOps solution accelerator is complete. See the next section for information on adapting this pattern to your use case and broader MLOps practices.

MLOps Next Steps

To adapt this pattern to your use case and code, a guide to modifying files and pipelines in the ML project repo is below:

/data

In the `/data` directory, you can place your data to be registered with Azure ML. A `data.yml` file describing the dataset is used by the training pipeline in `/mlops/azureml/train/` and should be modified as needed for your data. Likewise, the training pipeline should be modified to refer to your dataset.

/data-science/src

In this directory you will modify, add, or remove code for your data science workflow.

/data-science/environment

In this directory, modify `train-conda.yml` to define the python environment required by your model training.

/infrastructure

This directory contains the infrastructure template and infrastructure pipeline for your Azure ML environment. In general, it should not need modification but review by your IT and verification you can create the defined resources is recommended.

/mlops/azureml/train

This directory contains the Azure ML yaml definitions for your dataset (`data.yaml`), your python environment (`train-env.yaml`), and the Azure ML training pipeline itself (`pipeline.yaml`). Modify these as necessary to correctly refer to your data, environment, and python code steps as needed.

/mlops/azureml/deploy/batch and /mlops/azureml/deploy/online

These directories contain the yaml definitions and Azure ML pipelines for deployment of your model endpoints. Modify these as needed for your endpoint and model type.

/devops-pipelines

This directory contains the Azure DevOps pipeline definitions for deployment of Azure ML model training and endpoint pipelines. In general, these should need minimal changes except for updating references to data and training pipelines.

Troubleshooting

The following issues are commonly encountered on a fresh Azure DevOps organization.

"No hosted parallelism has been purchased or granted"

By default, new Azure DevOps organizations have **zero** parallel jobs on Microsoft-hosted agents. The infrastructure pipeline will queue indefinitely until a free grant is approved.

Resolution: Submit the free-tier parallelism request form: <https://aka.ms/azpipelines-parallelism-request>. Approval is typically granted within 2–3 business days. If you need to deploy immediately, configure a [self-hosted agent](#).

"TF402455: Pushes to this branch are not permitted" or pipelines never appear

The user account creating the pipelines (typically the project administrator running `initialise-project`) must have **Create build pipeline** permission on the project.

Resolution: In Azure DevOps, navigate to **Project Settings** → **Permissions** → **Contributors** (or your group), and ensure **Create build pipeline** is set to *Allow*.

"A pipeline with name `deploy-...` already exists" on retry

If the `initialise-project` pipeline fails partway through and you re-run it, the second run will fail because pipelines and folders from the first attempt still exist.

Resolution: Before retrying, delete:

1. The pipelines folder in **Pipelines** → **Pipelines** named after your repo.
2. The repository created in **Repos** for the project.
3. The corresponding `infrastructure/` folder if the infra pipeline partially executed and created resources in Azure — clean these up via the Azure portal or `az group delete`.

"No image label found to route agent pool Azure Pipelines. Pool: Azure Pipelines, Image: ubuntu-20.04"

`ubuntu-20.04` was retired from the Microsoft-hosted agent pool. This affects older versions of `mlops-project-template` cloned before May 2026.

Resolution: Update to the latest `mlops-project-template` (uses `ubuntu-latest`), or in your project repo replace `vmImage: ubuntu-20.04` with `vmImage: ubuntu-latest` in all `mlops/devops-pipelines/*.yaml` and `infrastructure/.../*.yaml` files.

`az pipelines create` fails with "Could not queue the build" during `initialise-project`

The default agent queue association may not be established on a brand-new project.

Resolution: The `create_ado_pipelines.sh` script now resolves the queue ID dynamically. If you use a self-hosted pool, set `AGENT_POOL_NAME` in the `initialise-project` pipeline variables before running.

Next Steps in MLOps

This guide illustrated using Azure DevOps pipelines and Azure Machine Learning pipelines to adopt training automation, deployment, and repeatability for your data science workflow for a single Azure ML environment. Follow on MLOps practices may include the following:

- By default, the Azure DevOps pipelines in this accelerator do not execute unless manually triggered. This is to avoid unnecessary automatic runs during initial deployment of the accelerator. However, you may want to modify the pipelines to enhance automation. A few examples:
 - Modify the deployed [Azure ML model training pipeline to run on a schedule](#)
 - Modify the manual trigger on the [Azure DevOps deploy-model-training-pipeline to trigger on a schedule](#)
 - Modify the Azure DevOps model deployment pipeline to [trigger upon successful completion of the model training pipeline](#) instead of a manual trigger
- Trunk-based development using development and test/staging branches in Azure DevOps. Development and Production environments are typically separated. You may create an additional Dev branch in your project repo and deploy a development Azure ML environment from that. When code and model development in that environment is satisfactory, code and pipeline changes can be merged into the main branch and updates deployed to Prod. A `config-infra-dev.yaml` example is provided in the project repo. To create this environment, repeat the steps in this guide from a dev branch in your ML project repo. Note that you should create a matching `dev` pipeline environment in Azure DevOps and new Azure service principal for an **Azure-ARM-Dev** service connection.
- Future work on this MLOps solution accelerator will include:
 - Integration of the [Azure Machine Learning registries](#) allowing you to register models and other artifacts from your development environment then deploying them in production directly.
 - Azure Machine Learning data drift monitoring
 - Integrated feature store
 - Deployment of secured Azure ML environments in a vnet

For questions, problems, or feature requests about this personal copy, please [submit an issue in the own repository](#).